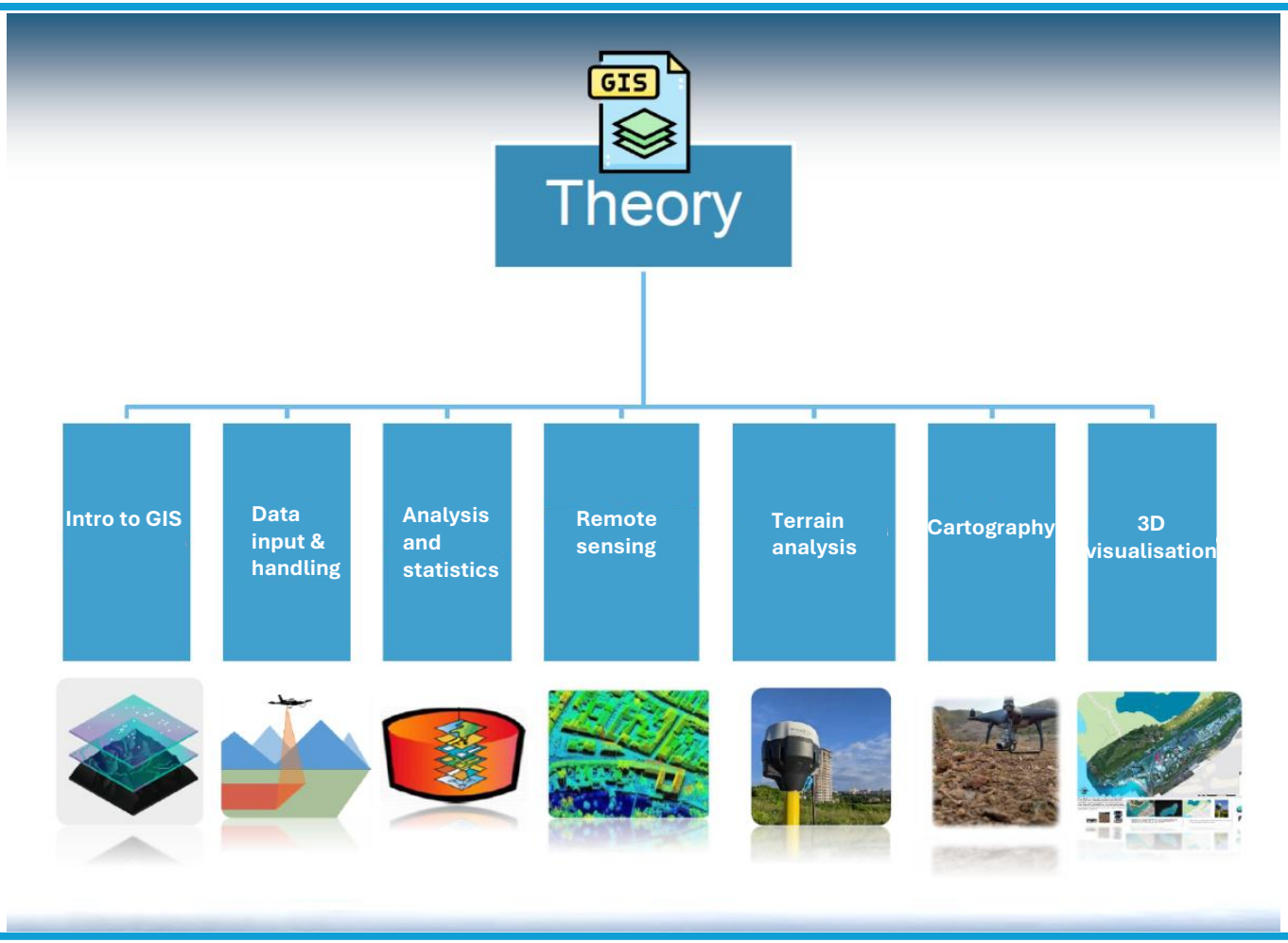
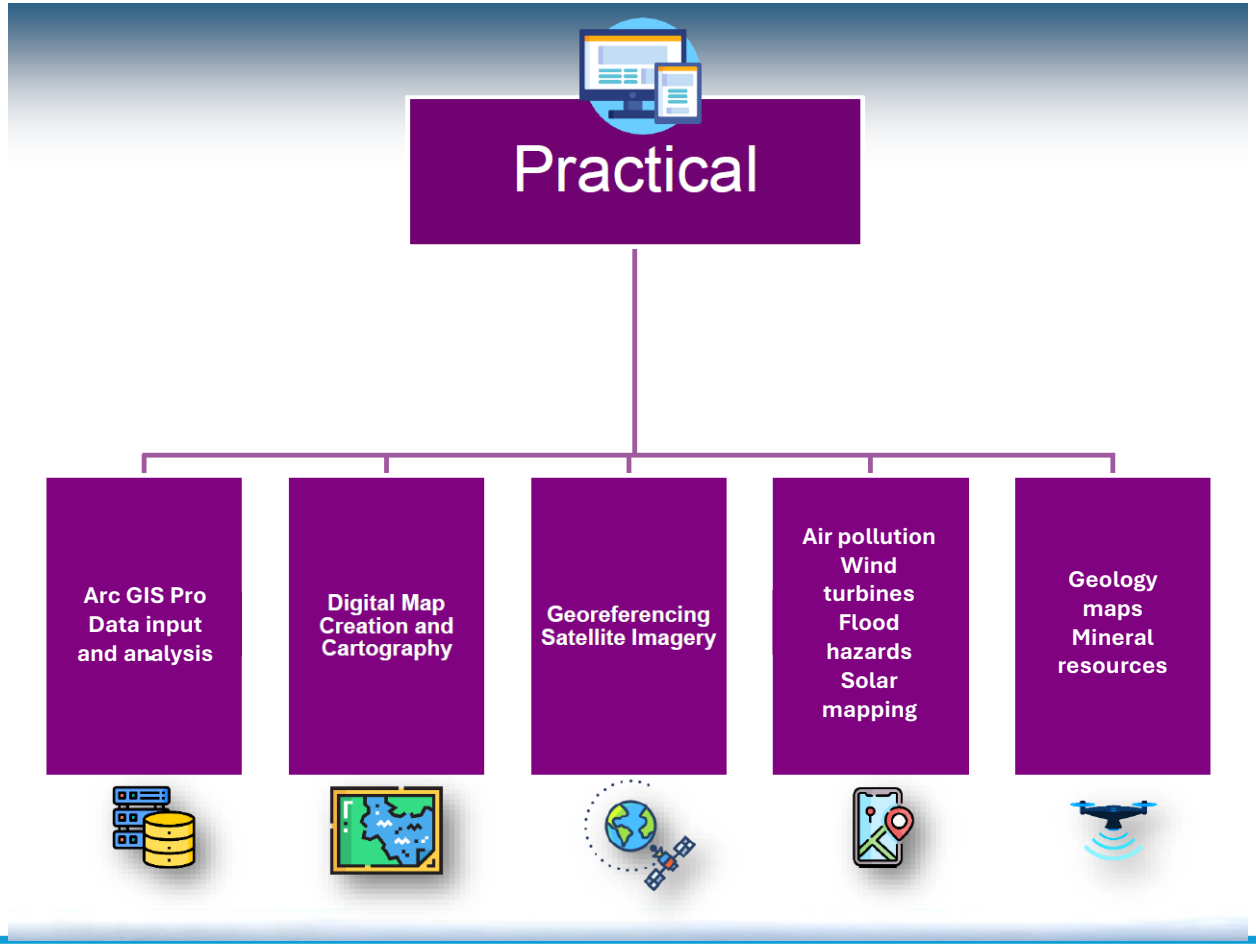
Course Overview - Aims

1. Introduce and develop understanding of the principles and applications of geospatial techniques.
2. Show how different types of geospatial data can be collected (including remote sensing techniques).
3. Develop skills required to deal effectively with geospatial data through real world examples.
4. To be able to add, store, analyse and present geospatial data using a Geographical information systems (GIS).
5. Develop proficiency in using the GIS software ArcGIS Pro.

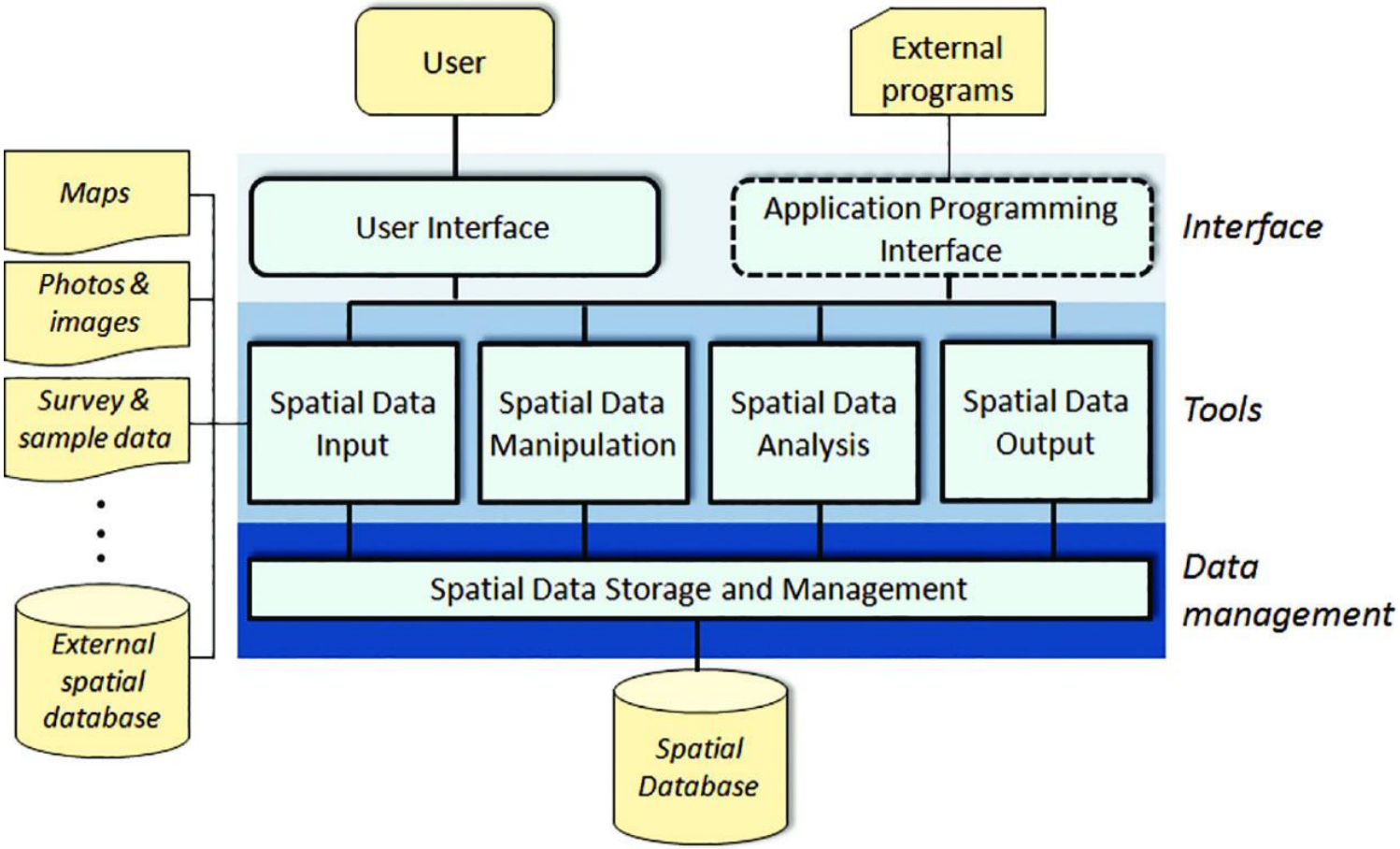
Theory


GIS
Theory
Intro to GIS
Data input & handling
Analysis and statistics
Remote sensing
Terrain analysis
Cartography
3D visualisation

Practical


Practical
Arc GIS Pro
Data input
and analysis
Digital Map
Creation and
Cartography
Georeferencing
Satellite Imagery
Air pollution
Wind
turbines
Flood
hazards
Solar
mapping
Geology
maps
Mineral
resources

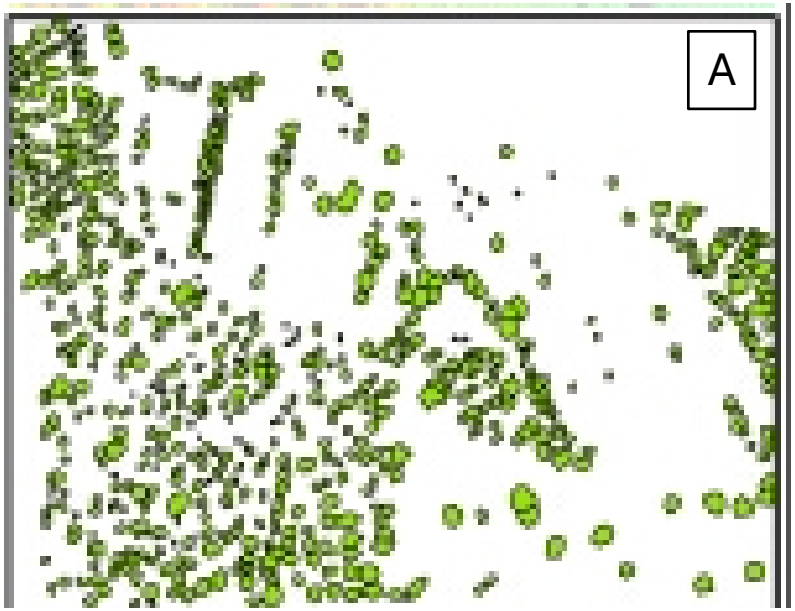
GIS functional components



Concept views of geographical space

The object view - point, line & area objects (vector).

The field view - continuous surfaces (raster).



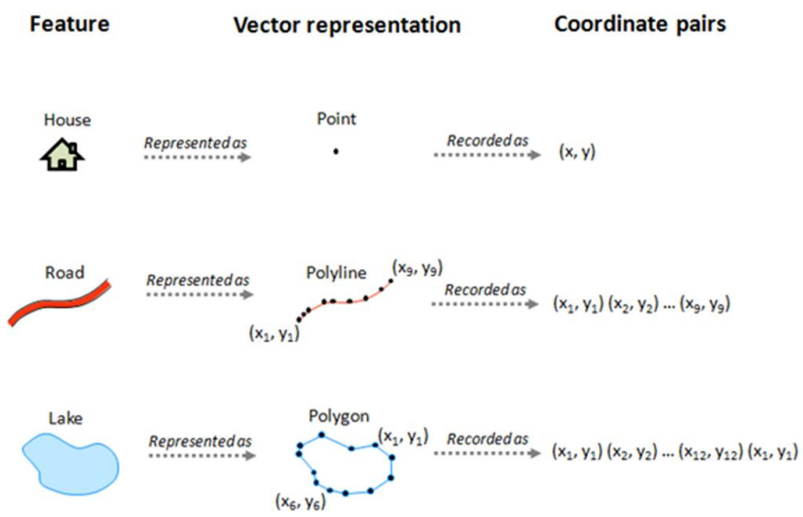
Trees as discrete objects



Tree height as continuous surface

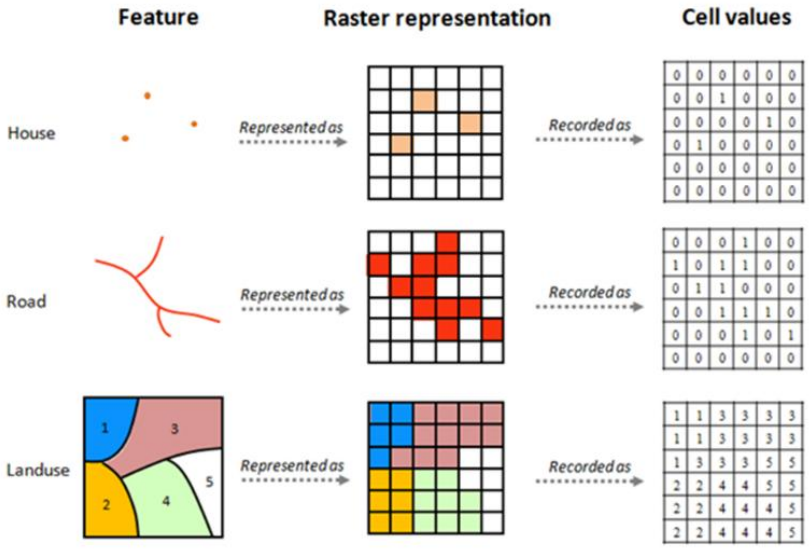
Vector data

Advantages	Disadvantages
Data can be represented in their original resolution without generalization.	More complex data structure.
Requires less disk storage space.	Inefficient for remotely sensed data.
Topological relationships are readily maintained.	Some spatial analysis procedures are complex and process intensive.
Graphical output resembles hand-drawn maps.	Overlaying multiple vector maps is often time consuming.

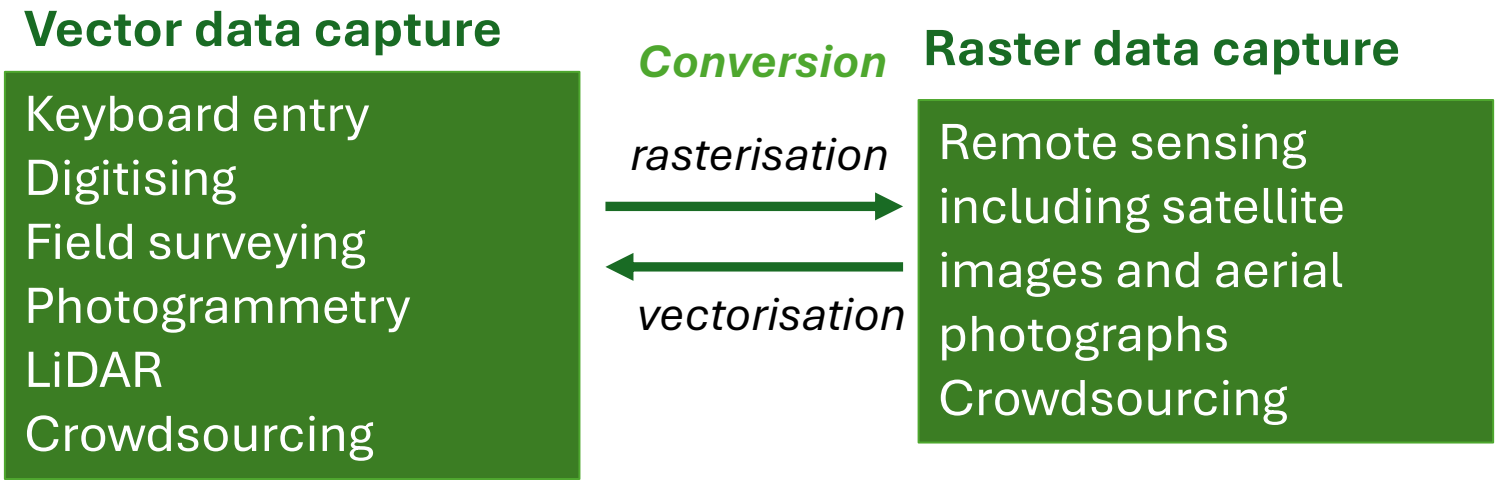


Raster data

Advantages	Disadvantages
Simple data structure.	Requires greater disk storage space.
Efficient for remotely sensed or scanned data (continuous data).	Depending on pixel size, graphical output may be less pleasing.
Simple spatial analysis procedures	Projection transformations are more difficult.
	Difficult to represent topological relationships.



Spatial data models



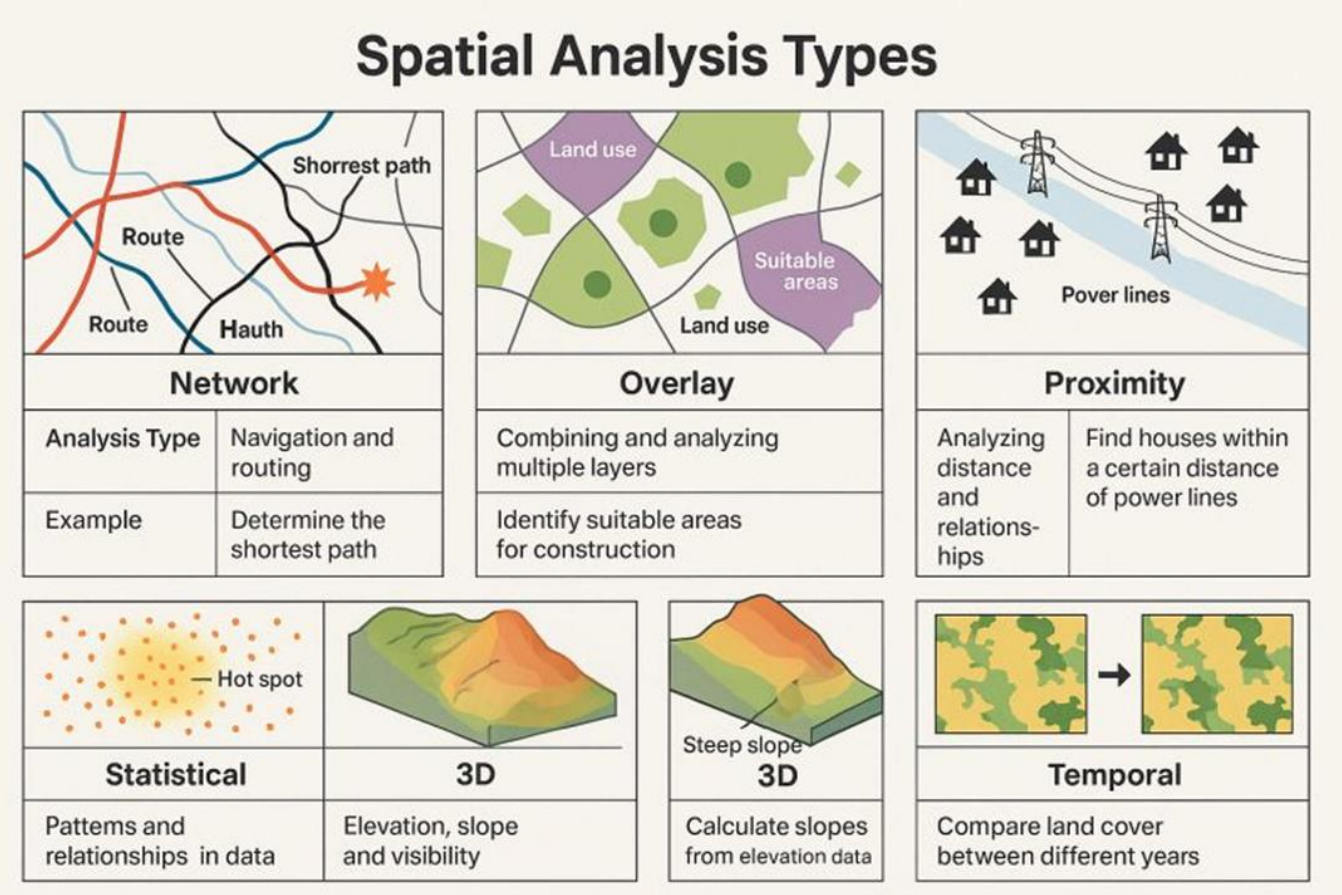
Map projections and coordinates systems

- Data are located on the Earth's surface using georeferences.
- The Earth's shape is modelled as a sphere (oblate spheroid).
- Spherical reference model e.g. WGS84 – often enhanced using an equipotential surface of the Earth's gravitational field at mean sea level (the geoid).
- Local datum takes account of regional variations e.g. NAD27, OSGB36.
- A geographical coordinate system defines a position on the Earth's surface in spherical coordinates (e.g. latitude and longitude).
- A projected coordinate system transforms the Earth's spherical surface to a 2D flat surface (maps, computer screen) and coordinates (x, y) are "flat" on a grid system e.g. UTM, British National Grid.

Global Navigation Satellite System (GNSS)

- A GNSS consists of a constellation of satellites orbiting the Earth.
- For global coverage, it is estimated that a constellation requires 18 to 30 satellites.
- GPS (USA), GLONASS (Russia), Galileo (Europe) and BeiDou (China) are four main GNSS systems that are currently operational.
- Position of an object is its latitude (distance from the equator), longitude (distance from the Greenwich meridian in the UK) and elevation above (or below) mean sea level.
- Four “visible” satellites are required for positioning with mm-scale accuracy possible.
- Hand-held GNSS receivers allow coordinates to be obtained at sample points.

Spatial analysis and modelling

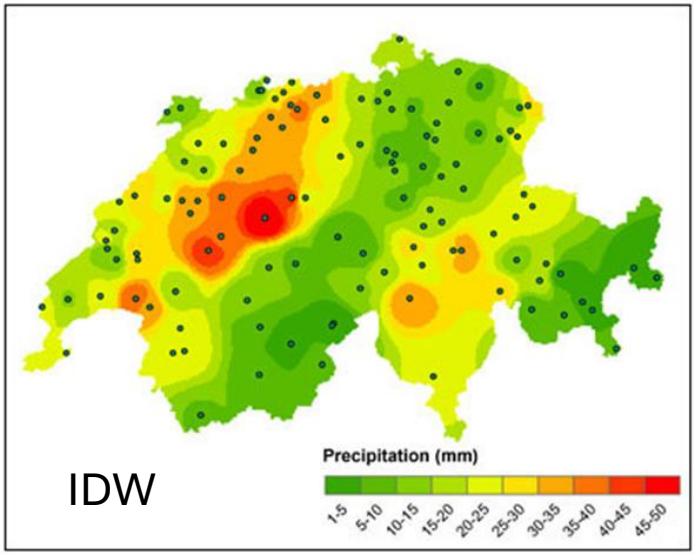


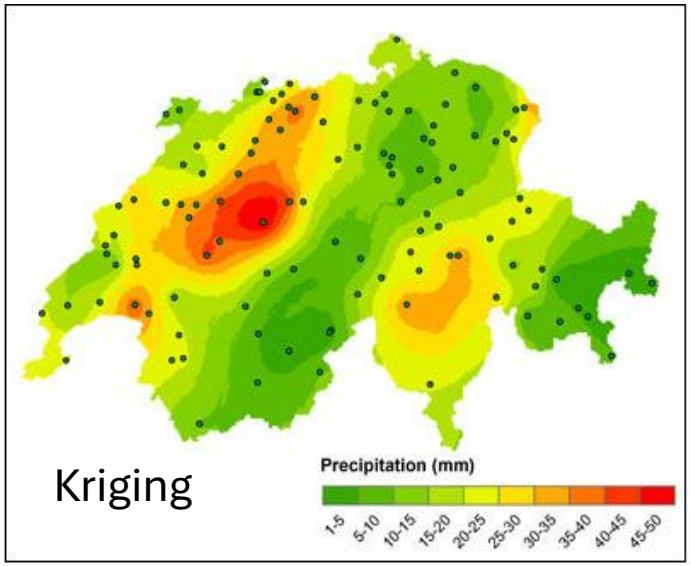
Methods of spatial analysis

- **Spatial query:** filtering and selecting data from a geodatabase based on attributes and location.
- **Spatial operations:** a type of spatial query class that includes operations such as intersection, containment, and distance calculations for objects in a given region or point.
- **Spatial interpolation:** estimates unknown values at specific locations based on known values from surrounding points.

Spatial interpolation methods

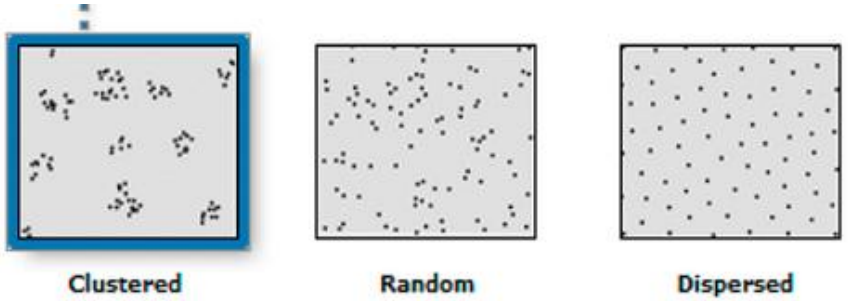
- Inverse distance weight (IDW) – deterministic (interpolated value only).
- Kriging – geostatistical (interpolated value and probability).





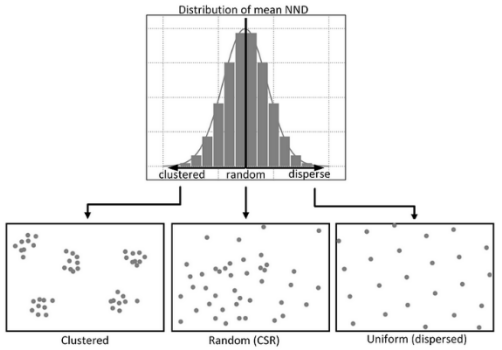
Spatial Autocorrelation

- Measure of the similarity (correlation) between nearby observations.
- Spatial pattern is clustered (close proximity), dispersed (spread out), or random.



Negative ----- spatial autocorrelation ----- Positive

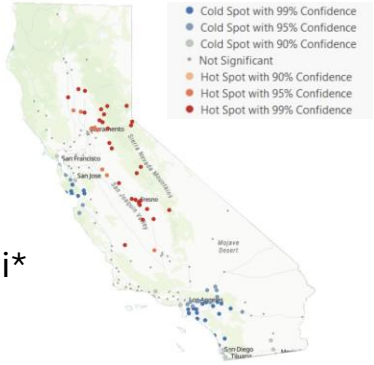
- Null hypothesis = spatial distribution is random.
- Alternate hypotheses = spatial clustered or dispersed.



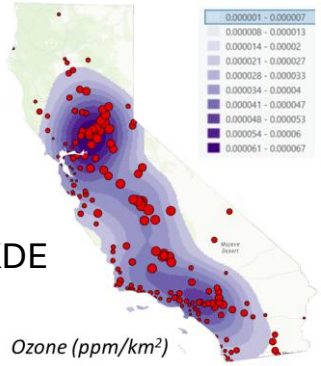
Spatial statistics

- *Nearest neighbour* – measures spatial autocorrelation using locations.
- *Moran's I* (local or global) - measures spatial auotocorrelation using feature values and their locations (a single statistic). Identifies clusters and outliers.
- *Getis Ord Gi** - hot-spot analysis identifies statistically significant clustering of high values (hot spots) and low values (cold spots).
- *Kernel density estimation* – shows heat maps as a smooth density surface across an area showing concentrations of locations or feature values).





Gi*

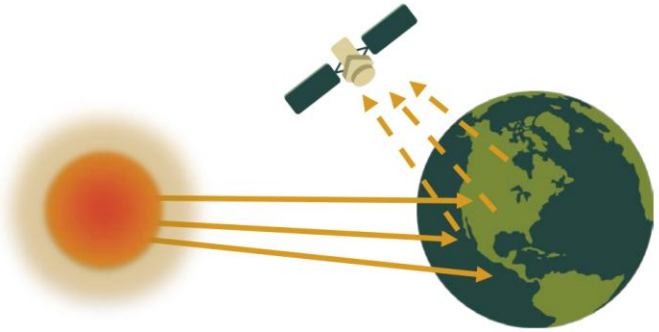


KDE

Remote sensing data

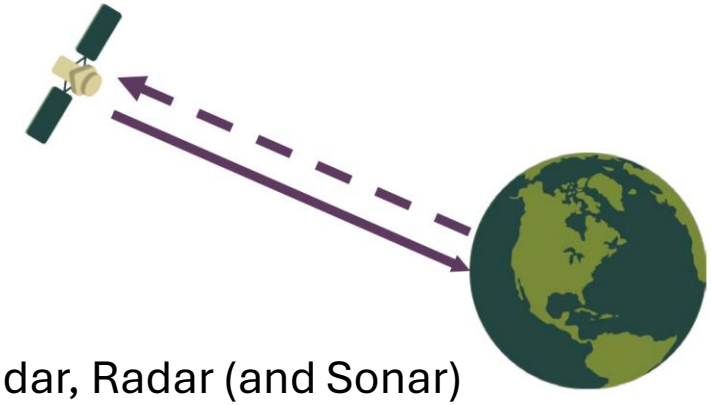
- Techniques for collecting environmental data from a distance without coming into direct contact with the objects of interest.
- Record the magnitude of electromagnetic energy flux that is reflected or emitted from the objects on the Earth's surface or within the atmosphere (normally imagery).
- Airplanes, satellites, and UAVs equipped with sensors.

Passive Sensors



Landsat and Sentinel 2 satellites

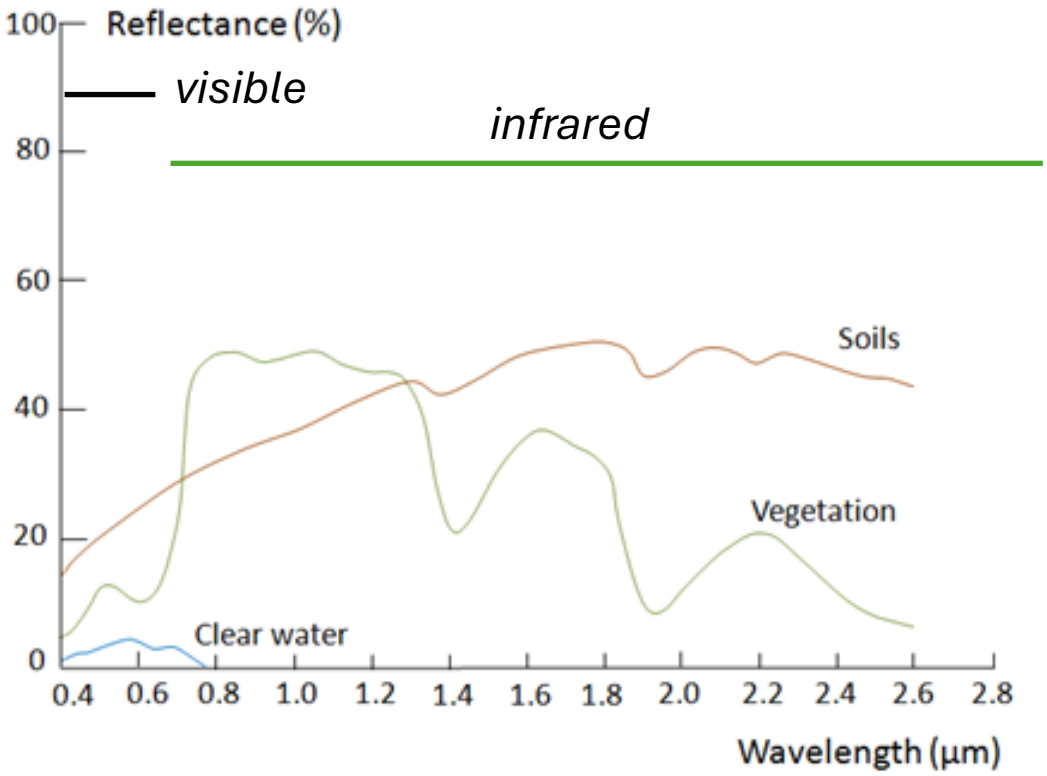
Active Sensors



Lidar, Radar (and Sonar)

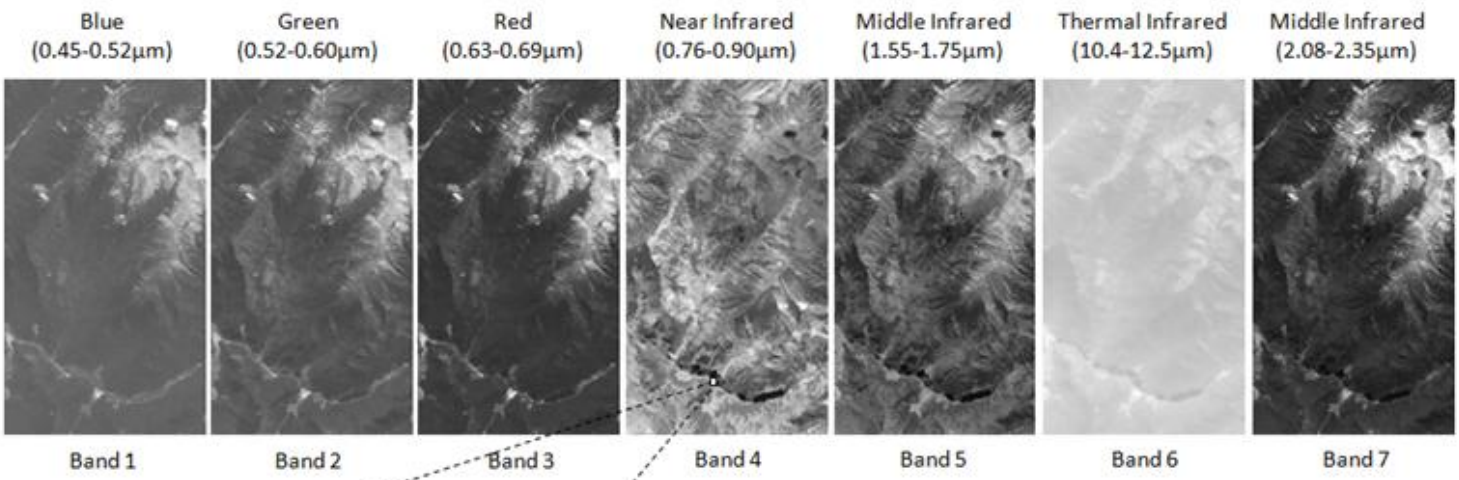
https://www.earthdata.nasa.gov/learn/backgrounders/remote-sensing

Spectral reflectance

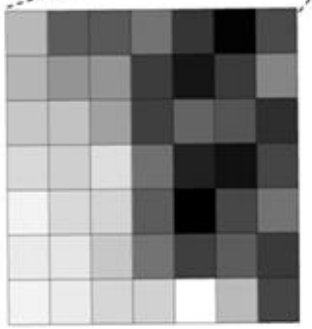


Digital multispectral images

Spectral band width or **spectral resolution**.



Pixel depth or
radiometric resolution
8-bit = 2^8 -1 = 0 -255.



A small portion of Band 4 image

93	48	47	59	31	7	40
91	75	75	34	15	33	68
99	97	81	35	52	45	27
108	103	110	58	21	13	38
120	108	106	48	4	39	59
116	114	99	59	35	49	33
120	117	107	105	126	94	38

Corresponding brightness values

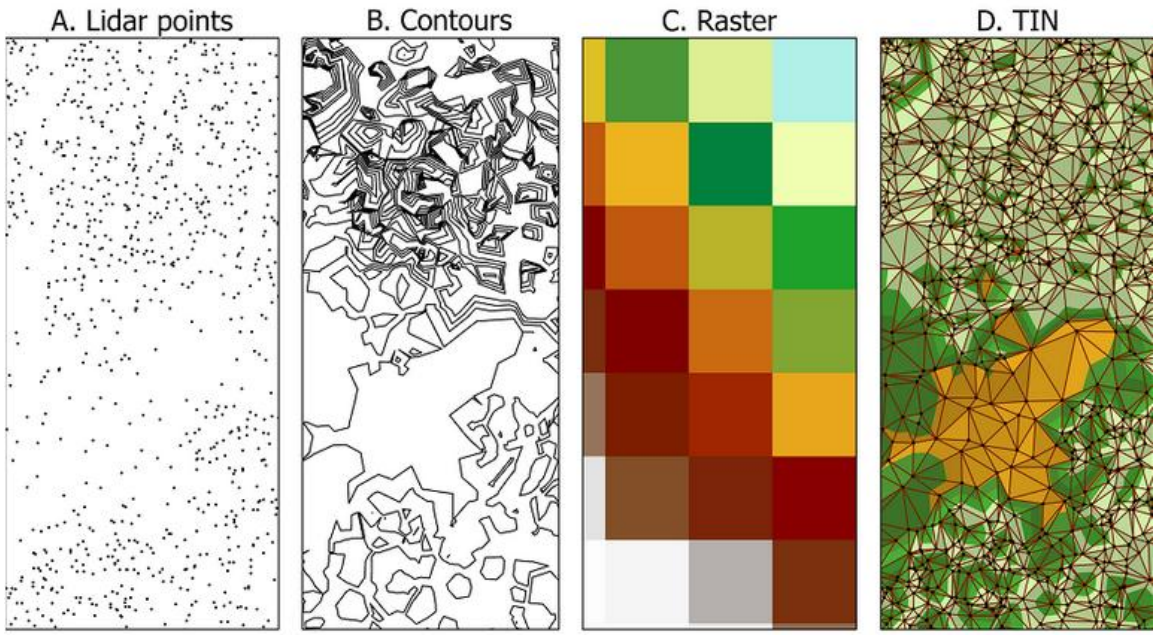
Pixel size or
spatial resolution

Sentinel spectral bands

Band number	Band name	Central wavelength (nm)	Spatial resolution (m)	
1	Coastal aerosol	442.7	60	aerosols, bathymetry
2	Blue	492.4	10	colour images
3	Green	559.8	10	
4	Red	664.6	10	vegetation health
5	Vegetation Red Edge	704.1	20	agriculture and forestry
6	Vegetation Red Edge	740.5	20	
7	Vegetation Red Edge	782.8	20	
8	NIR	832.8	10	vegetation, biomass, plant health
8A	Narrow NIR	864.7	20	
9	Water vapour	945.1	60	water vapour
10	SWIR – Cirrus	1373.5	60	cirrus clouds
11	SWIR	1613.7	20	burnt areas, moisture content
12	SWIR	2202.4	20	

Terrain analysis

- Contours, spot heights, DEM (raster) and TIN (vector).



different ways of representing the same continuous surface at the same scale and extent

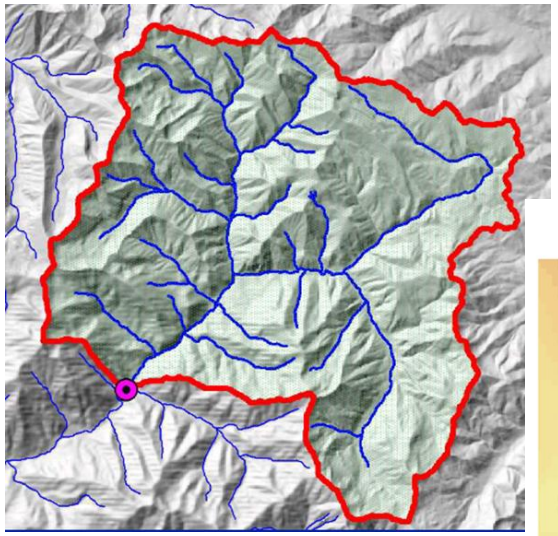
https://gistbok.ucgis.org/bok-topics/triangular-irregular-network-tin-models00

Primary and secondary terrain attributes

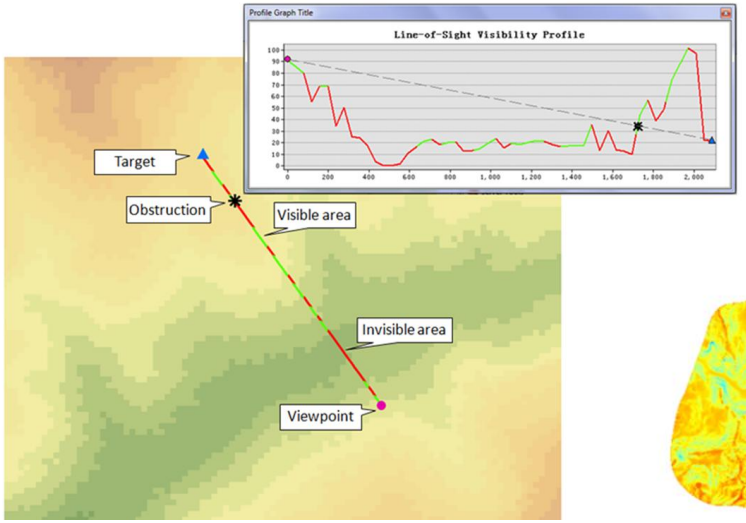
- Primary attributes are calculated directly from a DEM include elevation, slope, aspect, plan and profile curvature.
- Secondary attributes are computed from two or more primary attributes e.g. solar radiation, moisture index, sediment transport index, etc.
- Used to quantify the movement and distribution of water in the landscape, and the amount of solar radiation received at the surface.

Applications of terrain analysis

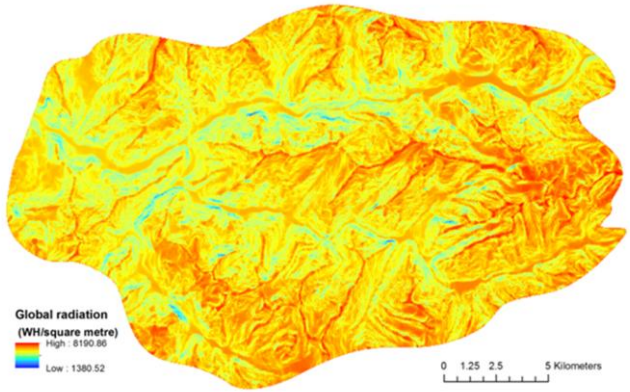
Catchment hydrology



Line of sight and viewsheds



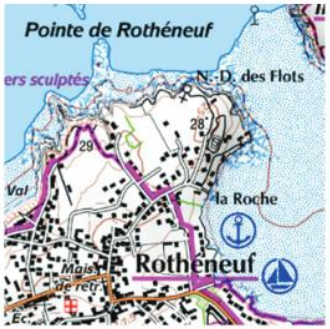
Solar radiation mapping



Cartographic editing and generalisation

Editing	Generalisation (cartographic abstraction)
Rubber sheeting – remove geometric distortions.	Selection – retain important features.
Transformation – rotate, scale, skew or translate features.	Simplification – remove complexity.
Edge matching – feature matching across layers.	Exaggeration – emphasise important features.
Line simplification – simplify while retaining shape.	Combination – merge similar features.
Error corrections - overshoot, unclosed polygon, etc.	Displacement – move to make distinguishable.
Reshaping individual features.	Classification – group similar features.

Large scale = small area

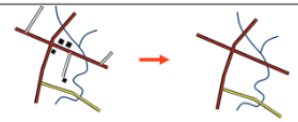
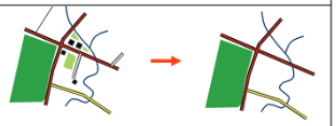
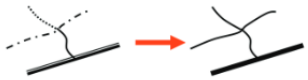
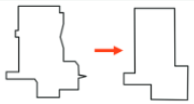
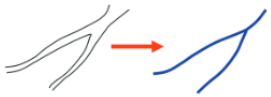
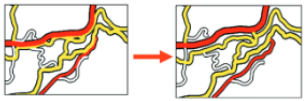
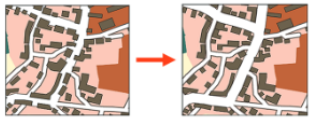








Small scale = large area

Operator	Description	Before → After	Before → After
Class selection	Select the classes of features that the model contains. Define the selection criteria for populating each class.		
Reclassification	Common or dominant characteristics of the objects are preserved. Subtle distinctions are ignored.		
Simplification	Important details are preserved, unimportant details are eliminated.		
Collapse	Reduction in the geometric dimension		
Enhancement	Make the shape of a feature more aesthetically pleasing.		
Elimination	When congestion occurs, unimportant objects are eliminated.		
Displacement	Important objects remain on their locations. Unimportant objects are moved away.		
Aggregation	Common boundaries, small gaps between neighbor objects are eliminated. General footprints are preserved.		
Combine	Regrouping a set of feature into a more abstract features, often of higher dimension		
Typification	Reduces the number of objects, preserves their distribution / pattern		

Mackaness et al (2017).. https://doi.org/10.1002/9781118786352.wbieg1015

Map types

General purpose maps

- provide geographical information for reference purposes
  - topographic maps - comprehensive reference information about a region
  - base map - a foundation for superimposing additional layers of information

Thematic maps

- map the spatial distribution of a single attribute or variable or the relationship among several variables

dot, isoline, choropleth, proportional, graduated, etc.

Map design

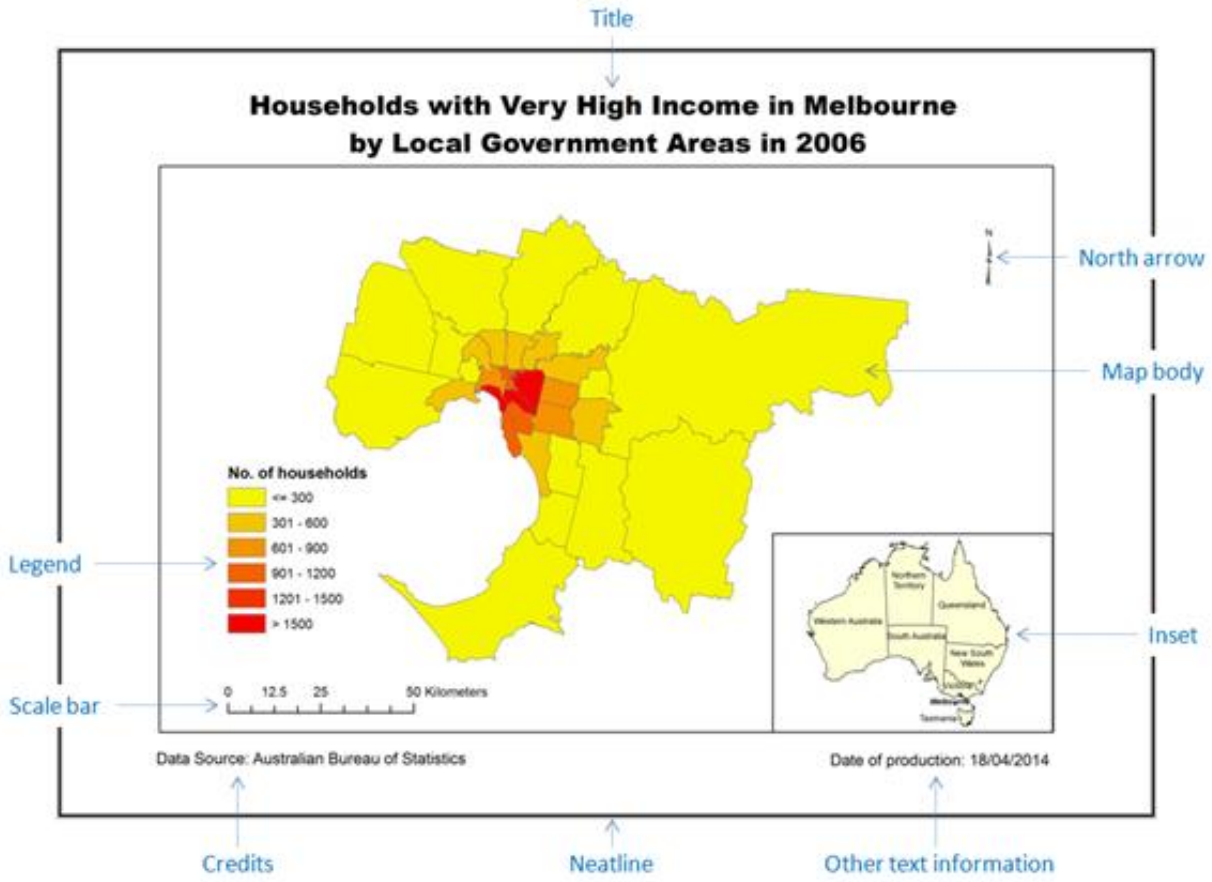
Selection: selecting the geographical area, geographical features, map scale, projection, data variables, etc.

Classification: grouping and categorising data

Symbolisation: selecting symbols and visual variables to represent data in a logical way

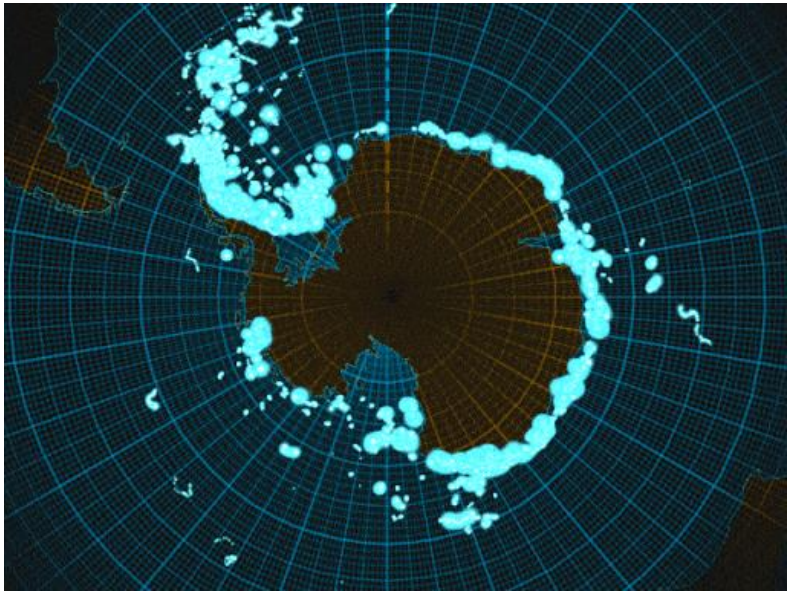
Layout design: organising disparate cartographic elements coherently on a page

Map elements



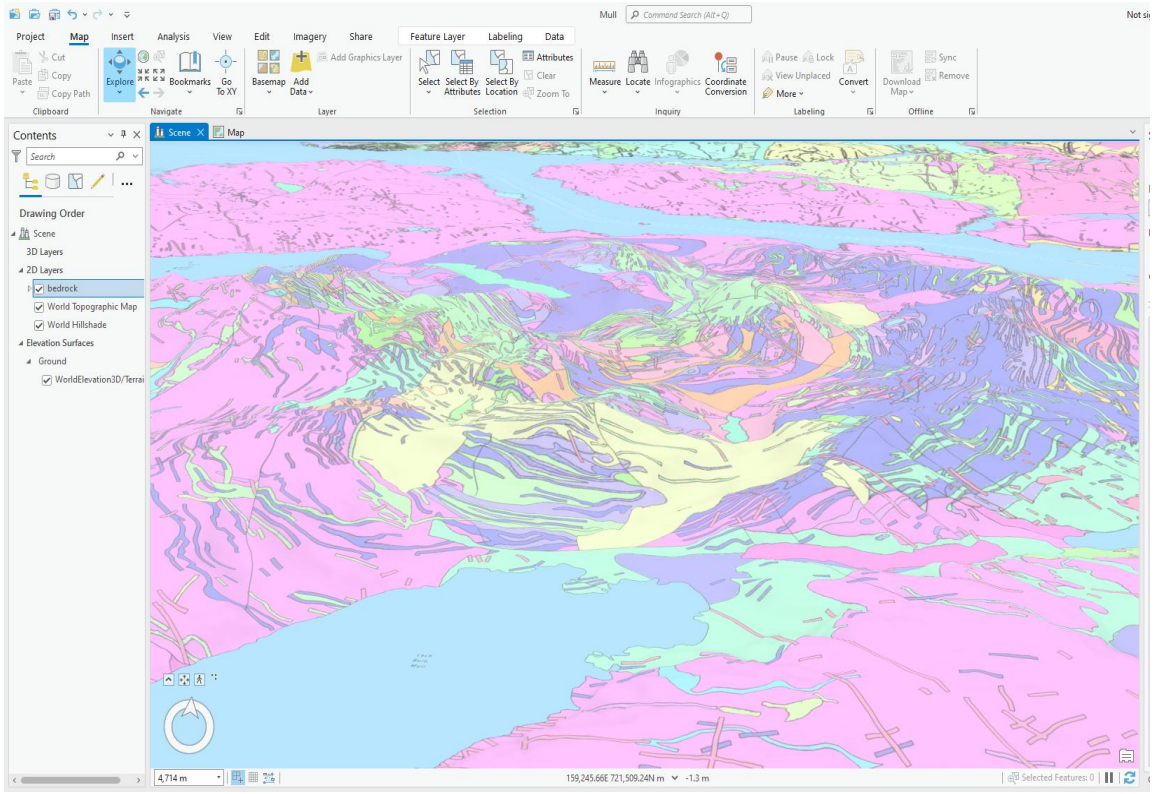
Animated mapping

- Animated mapping uses animation to show changes over **time, space** or **attributes**.
- Animations can be temporal and non-temporal.



animation showing the positions of Antarctic icebergs through time

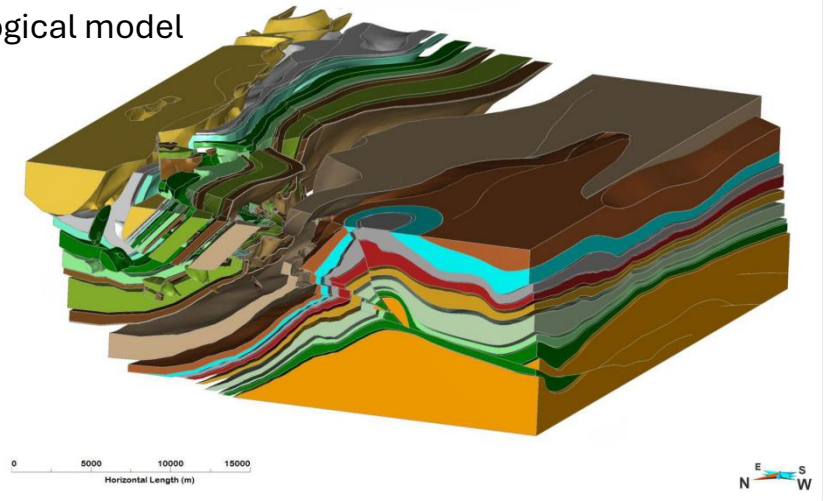
2.5D versus 3D mapping

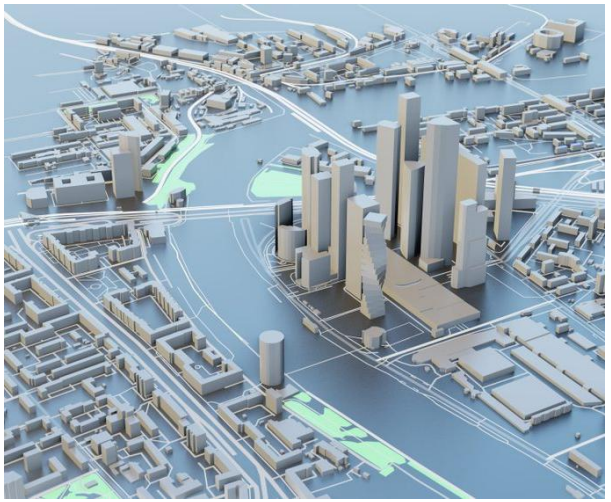


- Geological map draped over a DEM.
- The map is a surface existing in 3D space. But a point on this surface can still be described using only x and y coordinates (a 2 D surface).
- Third dimension represents a single attribute value e.g. elevation above sea level.
- Referred to as 2.5D maps.

True 3D

Geological model

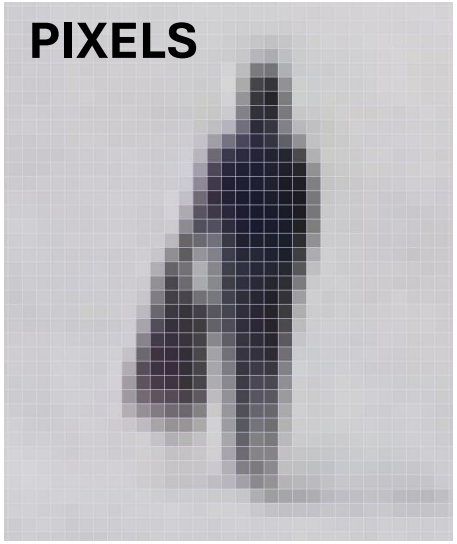




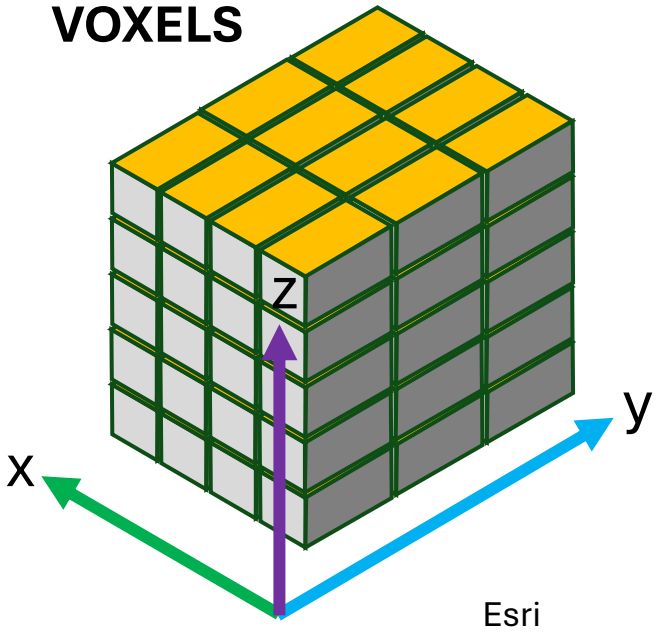
Digital twins

- Volume is integral in geospatial 3D.
- True 3D geospatial data requires three coordinates (x, y, z) because it represents volume, not just surface position.
- Without z, it is not possible to locate where inside the volume something is.

Digital space



The smallest unit of information in a 2D raster image. Each pixel has an x, y location.



A 3 dimensional pixel used to display 3D images.
A voxel = a small cube defined by x, y, z extents.

EART29102 Exam

- Online exam under controlled conditions i.e. is invigilated and not able to refer to notes, books, web resources or use AI.
- Exam is 70% of the final mark for the course
- 50 questions: majority are multiple choice, some are multiple answer.
- Multiple choice questions are similar in style to the lecture quiz questions.
- Use the weekly quizzes as practice.
- Take your time and read each question carefully.

ONE HOUR THIRTY MINUTES

THE UNIVERSITY OF MANCHESTER

GEOSPATIAL TECHNIQUES

08 June 2026

14:00 – 15:30

Instructions for candidates

Answer **ALL** questions in the online exam.

This is a closed book exam.

The use of AI software is prohibited.

There are 80 marks available on this exam. The number of marks for each question is shown and may be used as a guide to how much time you should spend on each.

Some questions require you to select **more than one correct answer** from a list.

- You **gain marks** for each correct answer selected.
- You **lose marks** for each incorrect answer selected.
- You **do not lose marks** for answers you do not select.
- The minimum score for each question is 0.

Answers to the online exam MUST be submitted at the end of the exam.

Additional permitted materials

The use of calculators is NOT permitted for this exam.

Example multiple answer question

Which TWO of the following are forms of geometric corrections applied to satellite multispectral images? *(2 marks)*

A. Removal of geometric distortions caused by sensor-Earth geometry variations.
B. Georeferencing an image to a particular projected map coordinate system.
C. Correcting an image for sensor response properties and viewing geometry.
D. Correcting an image for atmospheric effects and scene illumination.
E. Restoring an image to accurately represent the radiance measured by the sensor.

Answer: A and B are geometric corrections. The others are radiometric corrections. These corrections are applied before any image enhancement (e.g. band combinations) are applied.

No attempt	Select both correct answers	Select one correct answer only	Select any two, one correct, one incorrect	Select any two incorrect answers
0	2	1	0	0